

Do companies bend earnings to avoid reporting a loss?

A discontinuity test on 16,018 recent SEC firm-years, and why the answer is "not clearly"

The question

If reported earnings were unmanaged, the distribution of net income across thousands of firms should be smooth. There is no economic reason for a sharp break at exactly zero. But a manager staring at a small loss has discretion — the timing of a write-off, a slightly early revenue recognition, a reserve estimate nudged. If that discretion gets used to cross zero, the distribution should carry a **notch**: too few firms just below zero, too many just above.

Burgstahler and Dichev (1997) found exactly that notch and read it as evidence of earnings management to avoid losses. The finding is one of the most cited in empirical accounting.

Durtschi and Easton (2005) argued the inference does not follow. A discontinuity in a *scaled* earnings distribution can be produced by the scaling itself, by sample-selection, or by differences between the kinds of firms that fall either side of zero — none of which require a manager to do anything. Their title says it: the shapes of these distributions "are not evidence ipso facto."

That disagreement is what makes the exercise worth doing carefully. Replicating the notch is a small task. Finding out whether it survives contact with the critique is the real one.

Data and construction

The sample is built from the **SEC Financial Statement Data Sets** — quarterly flat files derived from XBRL filings, free and unauthenticated. I processed 37 quarterly archives spanning 2017q1 through 2026q1, scanning 113.2 million fact rows. The schema was verified against the live 2026q1 archive before any loader code was written; it has changed across the years and building against a remembered format would have been a silent source of error.

Only **10-K** filings are used, for a specific reason: a 10-K carries the comparative balance sheet, so one submission gives both current-year and prior-year total assets. That supplies the lagged denominator directly, from the filer's own comparative column, with no cross-filing join.

The measure is

$$ROA_{it} = \text{NetIncome}_{it} / \text{TotalAssets}_{i,t-1}$$

with **beginning-of-year** assets. Using end-of-year assets would put the current year's earnings into the denominator of the thing being measured — the most common way this analysis goes wrong.

Filters were fixed before any result was looked at, and every exclusion is counted in the README: 57,407 raw 10-K firm-years, less financial firms (14,590), utilities (1,404), missing industry codes, missing net income, non-positive lagged assets, and a \$10m size floor, leaves **31,240 firm-years across 5,366 firms**, fiscal years 2016–2025. Restricting to the analysis window $|ROA| \leq 0.10$ leaves **16,018 firm-years across 3,582 firms**. Firm-years are pooled, which inflates N relative to independent units — stated, not hidden.

Two things are enforced in code rather than by care. Zero always falls on a **bin boundary**: every edge is an integer multiple of the bin width, so a bin can never straddle the point being tested. And every test runs at **three** bin widths — 0.0025, 0.005, 0.01 — because a result that appears at only one width is a choice, not a finding.

Method

Each bin's count is predicted from its immediate neighbours, $\text{expected}_i = (\text{count}_{\{i-1\}} + \text{count}_{\{i+1\}}) / 2$, and the standardized difference uses the binomial variance standard in this literature. Both the bin immediately below zero and the bin immediately above are always reported: the hypothesis predicts a deficit below *and* a surplus above, and finding only one is weaker evidence.

What I found

The direction is right. The magnitude is not there.

Bin width	below zero	z	above zero	z
0.0025	246 obs / 252 pred	−0.31	291 / 252	+1.91
0.005	459 / 480	−0.79	550 / 496	+1.95
0.01	868 / 899	−0.87	1,083 / 1,017	+1.73

At every width the bin below zero is thinner than its neighbours predict and the bin above is thicker — the predicted pattern. But the deficit below zero is nowhere near significance ($p = 0.75, 0.43, 0.38$), and the surplus above sits right at the boundary ($p = 0.057, 0.051, 0.083$). A goodness-of-fit test that holds the two zero-adjacent bins out of a smooth fit does reject smoothness ($p \approx 0.001\text{--}0.005$), but the rejection is driven by the surplus above zero, not by any deficit below it.

The honest summary of the main test: **weak, directionally consistent, not significant**.

Then the robustness checks got interesting.

Scaling matters enormously. Rescaling the same firms by revenue rather than lagged assets produces a clear notch — $z = -2.97$ below zero ($p = 0.003$) and $+4.62$ above ($p < 0.001$), holding at all three bin widths. Book equity produces nothing ($z = -1.20$). Same firms, same years, same test; three scalars, three different answers, one of them publishable and two of them null. This is Durtschi and Easton's argument arriving empirically rather than theoretically, and it is the most useful thing in the project.

Size is not the story. The artifact explanation predicts the notch should concentrate among small firms, whose tiny denominators produce extreme scaled values. It does not: z below zero is -0.25 (small), -0.26 (medium), -0.91 (large). And the result is insensitive to the size floor across the full range from 0 to 500m, so the headline \$10m threshold is not doing any work.

The placebo is the punchline. Cash flow from operations, scaled by the same lagged assets, should be far harder to push across zero through accrual timing. The prediction is that the notch is absent there. Instead CFO shows a **larger break in the opposite direction** — a surplus just below zero and a deficit just above ($z = +3.55 / -3.59$ at width 0.01, $p < 0.001$).

That is not the simple scaling-artifact result, which would have pushed both measures the same way. It is something more awkward: the local-smoothness null fails, and fails significantly, on a measure that should not have a discontinuity at all. Note too that the CFO effect grows steadily with bin width ($+0.45 \rightarrow +2.04 \rightarrow +3.55$) — the signature of density curvature being read as a break, not of a knife-edge jump at zero. Whatever neighbour-averaging is picking up in CFO, it cannot be earnings management, and the same machinery is what produced the marginal z -statistics for net income.

What I conclude

I did not find evidence of a discontinuity at zero in scaled net income in recent SEC filing data. The direction of the effect matches the classical result at every bin width, which is worth something, but the deficit of small losses is not statistically significant under the headline specification; the one clearly significant version appears only under revenue scaling; and the placebo shows the test's own null assumption breaking down on data that cannot have been managed into shape.

Where a notch does appear, "consistent with earnings management" is as far as the evidence reaches. This test detects a distributional anomaly. It does not identify a manager, a decision, or an accrual, and it cannot separate management from any other mechanism that places extra mass just above zero.

Limitations, plainly

- **Pooled firm-years.** 5.8 years per firm on average; within-firm observations are not independent, so the p -values above are optimistic.
- **The null is the weak link.** Neighbour-averaging assumes local linearity in the density. The placebo demonstrates that assumption failing. A stronger design would model the density and test a break against the fitted model.
- **Period.** Fiscal years 2016–2025 only. Burgstahler and Dichev worked on 1976–1994. If the notch has genuinely faded in the post-SOX, post-XBRL era, a null here is compatible with their finding rather than a refutation of it. Testing that would require pushing the panel back before 2009, which the SEC data sets do not reach; the time-trend split was deliberately left out of scope.

- **Coverage.** XBRL-era filers only, so firms that deregistered before 2017 never appear. Revenue is tagged inconsistently across filers, so the revenue-scaled sample is a somewhat different set of firms than the asset-scaled one — which is itself a candidate explanation for why revenue scaling behaves differently.
- **Market value of equity** could not be tested at scale: the SEC flat files carry no price data. Book equity substituted on the main sample; market value was tested on a 237-observation yfinance sample, far too small to inform anything.

References

Burgstahler, D., & Dichev, I. (1997). Earnings management to avoid earnings decreases and losses. *Journal of Accounting and Economics*, 24(1), 99–126.

Durtschi, C., & Easton, P. (2005). Earnings Management? The Shapes of the Frequency Distributions of Earnings Metrics Are Not Evidence Ipso Facto. *Journal of Accounting Research*, 43(4), 557–592.

U.S. Securities and Exchange Commission. Financial Statement Data Sets. <https://www.sec.gov/data-research/sec-markets-data/financial-statement-data-sets>

Code, notebooks, the full exclusion log and all figures: see the repository README. Sample N = 16,018 firm-years (3,582 firms), fiscal years 2016–2025.